

Two Applications of Maximum Flow

1 The Bipartite Matching Problem

- a bipartite graph as a flow network
- maximum flow and maximum matching
- alternating paths
- certificates for bipartite graphs with no perfect matchings

2 Circulation with Demands

- flows with multiple sources and multiple sinks
- reduction to a flow problem
- circulations with demands and lower bounds

CS 401/MCS 401 Lecture 16
Computer Algorithms I
Jan Vershelde, 24 July 2026

Two Applications of Maximum Flow

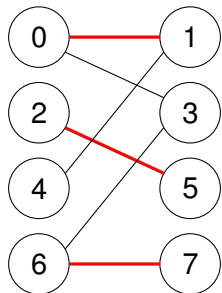
1 The Bipartite Matching Problem

- a bipartite graph as a flow network
- maximum flow and maximum matching
- alternating paths
- certificates for bipartite graphs with no perfect matchings

2 Circulation with Demands

- flows with multiple sources and multiple sinks
- reduction to a flow problem
- circulations with demands and lower bounds

the bipartite matching problem



In a bipartite graph $G = (V, E)$, we have:

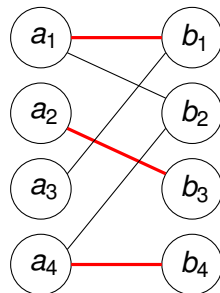
- 1 $V = A \cup B$, $A \cap B = \emptyset$, $A \neq \emptyset$, $B \neq \emptyset$.
- 2 for any edge $(a, b) \in E$: $a \in A$ and $b \in B$.

A **matching** M in a bipartite graph $G = (V, E)$:

- 1 $M \subseteq E$, is a subset of the set of edges,
- 2 every vertex appears at most once in M .

Given a bipartite graph G , the bipartite matching problem asks to find the matching M in G of the largest size.

assigning jobs



In $G = (V, E)$, $V = A \cup B$, $A \cap B = \emptyset$:

- 1 $A = \{a_1, a_2, a_3, a_4\}$ workers,
- 2 $B = \{b_1, b_2, b_3, b_4\}$ jobs, and
- 3 for any edge (a, b) : a can do job b .

A matching M in the bipartite graph G :

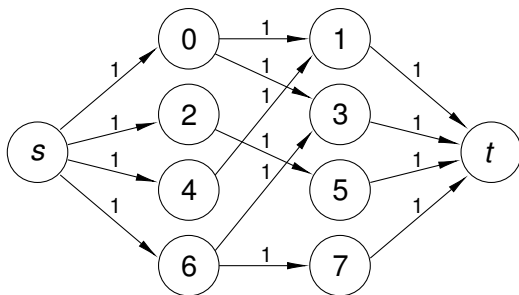
- 1 workers do only jobs they can do, and
- 2 every worker does at most one job.

We want to have as many jobs done as possible.

So, we ask for the matching M in G of the largest size.

the flow network for a bipartite graph

Given $G = (V, E)$, with $V = A \cup B$, add a source s and sink t , $V' = V \cup \{s, t\}$, and construct the flow network $G' = (V', E')$ as:



- 1 For all $e \in E$: $e \in E'$, all edges of G' are directed.
- 2 For all $a \in A$: $(s, a) \in E'$, connect the source to A .
- 3 For all $b \in B$: $(b, t) \in E'$, connect the sink to B .
- 4 For all edges $e \in E'$: $c_e = 1$, all capacities are one.

Two Applications of Maximum Flow

1 The Bipartite Matching Problem

- a bipartite graph as a flow network
- **maximum flow and maximum matching**
- alternating paths
- certificates for bipartite graphs with no perfect matchings

2 Circulation with Demands

- flows with multiple sources and multiple sinks
- reduction to a flow problem
- circulations with demands and lower bounds

matchings and flows

To a matching M in G corresponds a flow f in G' :

- Let $M = \{ (a_1, b_1), (a_2, b_2), \dots, (a_k, b_k) \}$.
- For all $(a, b) \in M$: $f(s, a) = 1$, $f(a, b) = 1$, and $f(b, t) = 1$.

The value of the flow is k , $v(f) = k$.

Exercise 1:

Verify that the f corresponding to a matching M is an s - t flow:

- 1 Show that the capacity conditions on f in G' hold.
- 2 Show that the conservation conditions on f in G' hold.

flows and matchings

If all capacities are integers, then there is an integer valued flow.

If we have an integer valued flow f in $G' = (V, E)$, then

$$M' = \{ (a, b) \in E' \mid f(a, b) = 1 \}.$$

Lemma (1)

If the value of the flow f is k , then $\#M' = k$.

Proof. G' is the flow network for the bipartite graph $G = (V, E)$, with $V = A \cup B$. Consider the cut which separates the source s from A . The value of the flow is $k = f^{\text{in}}(A)$. By the conservation conditions, $f^{\text{out}}(A) = f^{\text{in}}(A)$, the flow into A equals the flow out of A .

- The edges out of A that carry flow are the edges of M' .
- The flow at each edge equals one.

The number of edges with flow equals the size of M' .

Q.E.D.

the flow and its corresponding matching

If we have an integer valued flow f in $G' = (V, E)$, then for

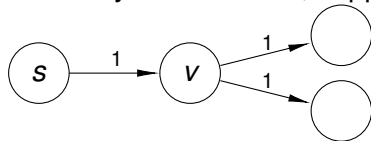
$$M' = \{ (a, b) \in E' \mid f(a, b) = 1 \},$$

we have to show that M' leads to a matching in $G = (V, E)$:

Lemma (2)

Every $v \in V$ appears at most once in M' .

Proof. By contradiction, suppose v appears twice in M' . If $v \in A$, then



This picture violates the conservation conditions at v , which contradicts that f is a flow in G' .

Thus v cannot appear twice in M' .

If $v \in B$, then a similar picture will contradict the conservation conditions of the flow.

Q.E.D.

maximum flow and maximum matching

Theorem (3)

Given a flow f in a flow network G' , constructed for a bipartite graph $G = (V, E)$, consider

$$M = \{ (a, b) \in E \mid f(a, b) = 1 \}.$$

Then, $v(f) = \#M$ and M is a matching in G .

Corollary (4)

If the value of the flow f in G' is maximal over all flows, then the matching M corresponding to f is maximal as well.

Exercise 2:

Write a proof for the corollary.

Two Applications of Maximum Flow

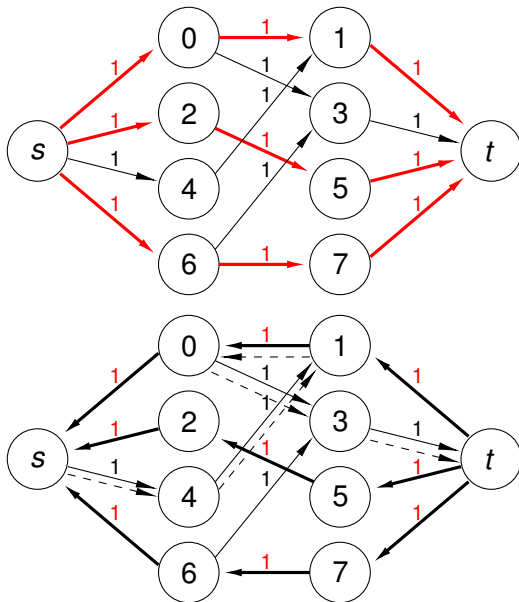
1 The Bipartite Matching Problem

- a bipartite graph as a flow network
- maximum flow and maximum matching
- **alternating paths**
- certificates for bipartite graphs with no perfect matchings

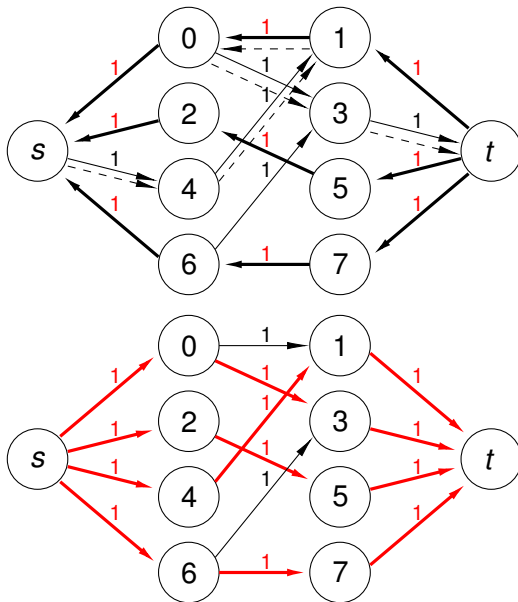
2 Circulation with Demands

- flows with multiple sources and multiple sinks
- reduction to a flow problem
- circulations with demands and lower bounds

finding an alternating path



the maximum bipartite matching



another exercise

When the Ford-Fulkerson algorithm is applied to the bipartite matching problem, the augmenting paths are called *alternating paths*, because those paths alternate between forward and backward edges when rerouting flow in the bipartite graph.

Exercise 3:

Consider the pictures on slides 12 and 13.

Describe the transitions between the top and the bottom graph, using the proper terminology introduced in lecture 15.

In particular, describe the construction of the residual graph with forward and backward edges in sufficient detail.

the running time

The Ford-Fulkerson algorithm runs in time $O(mC)$, $C = \sum_{e \text{ out of } s} c_e$,

for the flow network G' constructed from the bipartite graph $G = (V, E)$, $V = A \cup B$, $C \leq \#A \leq n$, where $n = \#V$, $m = \#E$.

Therefore, we arrive at the following theorem:

Theorem (5)

Consider a bipartite graph $G = (V, E)$, with $n = \#V$ and $m = \#E$. The maximum matching problem is solved by the Ford-Fulkerson algorithm in $O(mn)$ time.

Observe that we have a strongly polynomial time algorithm to solve the bipartite matching problem as the running time does not depend on the size of the input numbers, as for this problem, C is bounded by n .

solving a bipartite matching problem

Exercise 4:

Consider the bipartite graph $G = (V, E)$, defined by $V = X \cup Y$,

- $X = \{0, 2, 4, 6, 8\}$,
 - $Y = \{1, 3, 5, 7, 9\}$, and
 - $E = \{(0, 5), (0, 9), (2, 1), (2, 5), (2, 9), (4, 1), (4, 3), (4, 5), (6, 1), (6, 5), (6, 7), (8, 5), (8, 7), (8, 9)\}$.
- 1 Setup the flow network G' to apply the Ford-Fulkerson algorithm.
 - 2 Compute the maximum flow in G' .
 - 3 Explain how the maximum flow in the network corresponds to a bipartite matching, using G and G' in your explanation.

Two Applications of Maximum Flow

1 The Bipartite Matching Problem

- a bipartite graph as a flow network
- maximum flow and maximum matching
- alternating paths
- certificates for bipartite graphs with no perfect matchings

2 Circulation with Demands

- flows with multiple sources and multiple sinks
- reduction to a flow problem
- circulations with demands and lower bounds

perfect matchings

Definition (6)

Given a bipartite graph $G = (V, E)$ and a matching M .
 M is a *perfect* matching if every $v \in V$ appears in M .

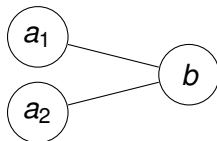
What if no perfect matching is found?

Consider a bipartite graph $G = (V, E)$, $V = A \cup B$.

If $a_1, a_2 \in A$:

- each incident to only one edge, and
- the same b is that end of their incident edge.

Either a_1 or a_2 is excluded from the matching.



a certificate for no perfect matching?

How can we be sure no perfect matching exists?

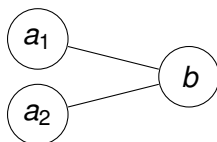
Is there a certificate?

We generalize the example on the previous slide.

Definition (7)

Let $G = (V, E)$ be a bipartite graph, $V = A \cup B$. For any subset $S \subseteq A$, define $\Gamma(S) \subseteq B$ as: $\Gamma(S) = \{ b \in B \mid (a, b) \in E, a \in S \}$.

In the example



consider $S = \{ a_1, a_2 \}$, then $\Gamma(S) = \{ b \}$ and $\#\Gamma(S) = 1 < 2 = \#S$.

a certificate for a perfect matching

Let $G = (V, E)$ be a bipartite graph, $V = A \cup B$, $A \cap B = \emptyset$.

For any subset $S \subseteq A$,

$$\Gamma(S) = \{ b \in B \mid (a, b) \in E, a \in S \}$$

is the set of all vertices in B connected to a vertex in S .

If we have a perfect matching, then for any subset S of A , we have as many vertices in B connected to vertices in S .

Proposition (8)

If there is a perfect matching in a bipartite graph $G = (V, E)$ with $V = A \cup B$, then for all $S \subseteq A$: $\#\Gamma(S) \geq \#S$.

By contraposition: if there is a $S \subseteq A$: $\#\Gamma(S) < \#S$, then G has no perfect matching. Does the converse hold?

Hall's theorem

Theorem (9)

Let $G = (V, E)$ be a bipartite graph, $V = A \cup B$ with $\#A = \#B$. Then,

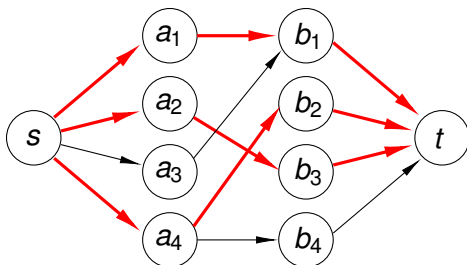
- either G has a perfect matching,
- or there is a $S \subseteq A$: $\#\Gamma(S) < \#S$.

A perfect matching or a certificate subset S can be found in $O(mn)$ time, where $n = \#V$ and $m = \#E$.

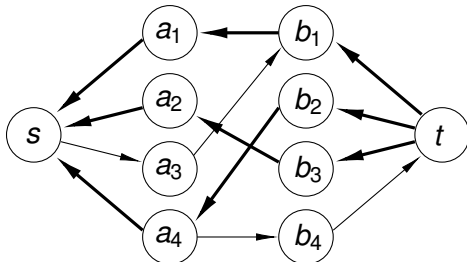
Outline of the proof, via an algorithm to compute the certificate:

- 1 The Ford-Fulkerson algorithm gives the maximum flow in $O(mn)$.
- 2 The value of the maximum flow equals the capacity of the minimum cut.
- 3 A breadth-first or depth-first search computes the cut in $O(m)$.
- 4 The minimum cut can be modified to find $S \subseteq A$: $\#\Gamma(S) < \#S$.

a matching that is not perfect

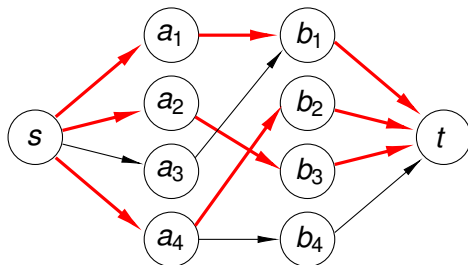


The residual graph:



The cut $X = \{s, a_3, b_1, a_1\}$, $Y = V \setminus X = \{a_2, a_4, b_2, b_3, b_4, t\}$.

computing the certificate from the cut



The cut is defined by $X = \{s, a_3, b_1, a_1\}$ and $Y = \{a_2, a_4, b_2, b_3, b_4, t\}$.

Take $S = \{a_1, a_3\} = X \cap A$. Then $\Gamma(S) = \{b_1\}$, and

$$\#\Gamma(S) = 1 < 2 = \#S.$$

defining the certificate

Proposition (10)

Let the graph $G = (V, E)$, with $V = A \cup B$,

- be bipartite with no perfect matching, and
- X and Y define the minimal cut in the flow network.

Then, for $S = X \cap A$, $\# \Gamma(S) < \# S$.

Proof. Let M be the matching with the flow equal to the cut.

S contains at least one vertex a_i that is not in the matching M .

- 1 If there is no $(a_i, b) \in E$, for some $b \in B$, then $\# \Gamma(S) < \# S$.
- 2 Otherwise, if, for $j \neq i$: $(a_j, b) \in M$, then $\# \Gamma(S) < \# S$, because $a_j \in S$ is implied by $(a_j, b) \in M$.

If there is no $j \neq i$: $(a_j, b) \notin M$, then the flow is not optimal, for adding (a_i, b) to M would increase the flow by one.

Q.E.D.

assigning multiple jobs to worker

As an application of bipartite matching, we assigned at most one job to each worker. What if workers can do more than one job?

Exercise 5:

Consider the job assignment problem in a bipartite graph $G = (V, E)$, $V = A \cup B$, $A \cap B = \emptyset$, where

- the set A represents m workers,
- the set B represents n jobs, and
- if, for $a \in A$ and $b \in B$, $(a, b) \in E$, then worker a can do job b .

Define the flow network if every worker can do up to three jobs.

Does the Ford-Fulkerson algorithm apply to this problem?

Two Applications of Maximum Flow

1 The Bipartite Matching Problem

- a bipartite graph as a flow network
- maximum flow and maximum matching
- alternating paths
- certificates for bipartite graphs with no perfect matchings

2 Circulation with Demands

- flows with multiple sources and multiple sinks
- reduction to a flow problem
- circulations with demands and lower bounds

CS 401/MCS 401 Lecture 16
Computer Algorithms I
Jan Verschelde, 24 July 2026

Two Applications of Maximum Flow

1 The Bipartite Matching Problem

- a bipartite graph as a flow network
- maximum flow and maximum matching
- alternating paths
- certificates for bipartite graphs with no perfect matchings

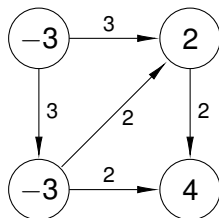
2 Circulation with Demands

- flows with multiple sources and multiple sinks
- reduction to a flow problem
- circulations with demands and lower bounds

supplies and demands

Consider a flow network with multiple sources and multiple sinks.

- the sources represent factories who provide supply,
- the sinks represent retail outlets who have demand.



Given a flow network $G = (V, E)$ with integer capacities c_e on the edges.

To each vertex v corresponds a demand d_v :

- if $d_v > 0$, v has demand of d_v and is a sink,
- if $d_v < 0$, v has supply of d_v and is a source,
- if $d_v = 0$, v is an internal node.

In our running example, the label of each vertex v is d_v , and the values of the demands d_v are inside the circle for each vertex.

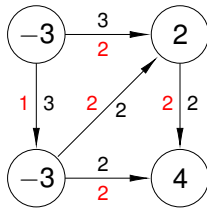
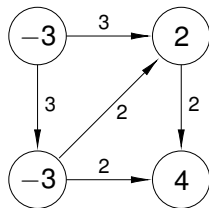
The capacity $c_{(u,v)}$ on the edge (u, v) represents how many units can be shipped from u to v .

circulations and a feasibility problem

A **circulation** with demands $\{d_v\}$ is a function $f : E \rightarrow \mathbb{R}^+$ that assigns to each edge a nonnegative number, subject to

- *capacity conditions*: for each $e \in E$: $0 \leq f(e) \leq c_e$.
- *conservation conditions*: for each $v \in V$: $f^{\text{in}}(v) - f^{\text{out}}(v) = d_v$.

Instead of an optimization problem, we are interested in **feasibility**: does there exist a circulation f which satisfies the capacity and conservation conditions?



To verify $d_v = f^{\text{in}}(v) - f^{\text{out}}(v)$, add the capacities of the incoming edges, and subtract the capacities of the outgoing edges.

supply equals demand

Lemma (10)

If there is a feasible circulation in a graph $G = (V, E)$ with demands $\{d_v\}$, then $\sum_{v \in V} d_v = 0$.

Proof. In a feasible solution f , the conservation conditions are satisfied: for each $v \in V$: $f^{\text{in}}(v) - f^{\text{out}}(v) = d_v$. Summing over all v :

$$\sum_{v \in V} d_v = \sum_{v \in V} f^{\text{in}}(v) - \sum_{v \in V} f^{\text{out}}(v).$$

Consider the edge $(u, v) \in E$. Along (u, v) , what goes out a vertex is $f^{\text{out}}(u)$ and what comes in is $f^{\text{in}}(v)$. Summing over all vertices,

- the term $f^{\text{in}}(v)$ is canceled by $f^{\text{out}}(v)$ in the other sum, and
- the term $f^{\text{out}}(u)$ is canceled by $f^{\text{in}}(u)$ in the other sum.

So the total sum equals zero.

Q.E.D.

the capacity D

Corollary (11)

For a feasible circulation with demands $\{d_v\}$:

$$\sum_{v \in V: d_v > 0} d_v = \sum_{v \in V: d_v < 0} -d_v.$$

The common value is denoted by

$$D = \sum_{v \in V: d_v > 0} d_v = \sum_{v \in V: d_v < 0} -d_v.$$

Two Applications of Maximum Flow

1 The Bipartite Matching Problem

- a bipartite graph as a flow network
- maximum flow and maximum matching
- alternating paths
- certificates for bipartite graphs with no perfect matchings

2 Circulation with Demands

- flows with multiple sources and multiple sinks
- **reduction to a flow problem**
- circulations with demands and lower bounds

reduction to a flow problem

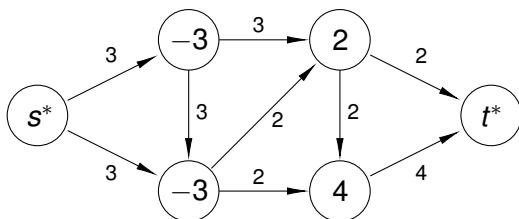
- Let S denote the set of all vertices v with $d_v < 0$.
- Let T denote the set of all vertices v with $d_v > 0$.

Attach a super source s^* to each vertex $u \in S$:

- for each $u \in S$, add the edge (s^*, u) with capacity $-d_u$.

Attach a super sink t^* to each vertex $v \in T$:

- for each $v \in T$, add the edge (v, t^*) with capacity d_v .



In this graph G' , we search for a maximum s^*-t^* flow.

an exercise

Exercise 6:

Run the Ford-Fulkerson algorithm on the flow network shown on the previous slide.

Show all steps in the execution of the algorithm.

circulation and maximum flow

Theorem (12)

*Let $G = (V, E)$ be a flow network with demands $\{d_v\}$.
Let G' be the graph with added source s^* and sink t^* .
There is a feasible circulation if and only if
the maximum s^*-t^* flow in G' has value D .*

Proof. We cannot have an s^*-t^* flow in G' of value larger than D because the cut (A, B) with $A = \{s^*\}$ has only capacity D .

- ① $\Rightarrow$: A feasible circulation f sends
- ▶ a flow value of $-d_v$ on each edge (s^*, v) , and
 - ▶ a flow value d_v on each edge (v, t^*) .

Summing over all vertices, we obtain an s^*-t^* flow of value D , which is maximal.

circulation and maximum flow continued

Theorem (12)

*Let $G = (V, E)$ be a flow network with demands $\{d_v\}$.
Let G' be the graph with added source s^* and sink t^* .
There is a feasible circulation if and only if
the maximum $s^* - t^*$ flow in G' has value D .*

The proof continues with the other direction.

② $\Leftarrow$: Consider a maximum $s^* - t^*$ flow in G' of value D .

Then, every edge out of s^* and every edge into t^* is completely saturated with flow.

If we delete those edges, we obtain a circulation f in G which satisfies $f^{\text{in}}(v) - f^{\text{out}}(v) = d_v$, for every $v \in V$, so the conservation conditions are met.

Q.E.D.

running time

Theorem (13)

*Let $G = (V, E)$ be a flow network with demands $\{d_v\}$.
Let $G' = (V', E')$ be the graph with added source s^* and sink t^* .
If all capacities and demands are integer numbers,
then the feasible circulation has an integer value
and can be computed in $O(mD)$ time, where $m = \#E'$.*

The theorem follows from the application of the Ford-Fulkerson algorithm and the observation that the constant $C = \sum_{e \text{ out of } s} c_e$ equals D in the graph G' .

Observe that we have a pseudo polynomial time algorithm to solve the feasibility problem as the running time does depend on the size of the input numbers, as for this problem, C equals D .

a transportation problem

Exercise 7:

Consider a network with four nodes:

- nodes 1 and 2 have supply 12 and 11, respectively, and
- nodes 3 and 4 have demand 19 and 4, respectively.

The capacities on the edges between the nodes are

$$c_{(1,3)} = 12, c_{(2,1)} = 4, c_{(2,4)} = 14, c_{(3,2)} = 9, c_{(4,3)} = 7.$$

Is there a feasible circulation?

Justify your answer via the reduction to a maximum flow problem.

Two Applications of Maximum Flow

1 The Bipartite Matching Problem

- a bipartite graph as a flow network
- maximum flow and maximum matching
- alternating paths
- certificates for bipartite graphs with no perfect matchings

2 Circulation with Demands

- flows with multiple sources and multiple sinks
- reduction to a flow problem
- circulations with demands and lower bounds

circulations with demands and lower bounds

Placing lower bounds on edges forces flow to make use of edges.

Consider a flow network $G = (V, E)$, with for each $e \in E$:

- ℓ_e is *the lower bound* on e , $\ell_e \geq 0$,
- c_e is the capacity of e , $c_e \geq \ell_e$.

For every $v \in V$, d_v is the demand at v , d_v can be positive or negative.

We assume that all ℓ_e , c_e , and d_v are integer numbers.

A feasible circulation f in the flow network G satisfies

- 1 *the capacity conditions*: for all $e \in E$: $\ell_e \leq f(e) \leq c_e$, and
- 2 *the conservation conditions*: for all $v \in V$: $f^{\text{in}}(v) - f^{\text{out}}(v) = d_v$.

an application: course planning

Exercise 8: At the start of each semester, students enroll in courses.

Courses are represented by nodes and directed edges between the nodes represent the prerequisites. Supply and demand at nodes:

- supply equals the number of students with credit in a course,
- demand equals the number of students interested in a course.

Capacities on edges have lower and upper bounds:

- lower bound: minimum number of students to run a course,
- upper bound: number of available seats in a classroom.

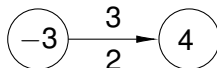
The problem is to decide for a given set of course offerings and a pool of students, whether the supply can meet demand, subject to all constraints of prerequisites and capacities.

Give a nontrivial numerical example of this problem.

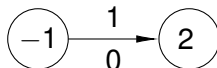
Explain how to compute a feasible solution to this problem.

eliminating lower bounds on edges

Consider an edge with lower bound 2 and capacity 3 between a node with demand -3 (supply of 3) and demand 4.



If we send a flow of 2 along the edge, then



- 1 the lower bound is met, we set it to zero,
- 2 we can reduce the upper bound by 2, it becomes 1,
- 3 the supply in the left node drops by 2, it becomes -1 ,
- 4 the demand in the right node drops by 2, it becomes 2.

constructing a flow network without bounds

Given a flow network $G = (V, E)$, with ℓ_e , c_e , and d_v .

Consider an initial flow f_0 on G : $f_0(e) = \ell_e$.

Construct $G' = (V', E')$, with the same vertices and edges, but for $e' \in E'$ with corresponding edge $e \in E$:

- ① $\ell_{e'} = 0$, met by the initial flow f_0 ,
- ② $c_{e'} = c_e - \ell_e$, reduced capacity,
- ③ $d_v = ?$

For the initial flow f_0 , for all $v \in V$:

$$f_0^{\text{in}}(v) - f_0^{\text{out}}(v) = \sum_{e \text{ into } v} \ell_e - \sum_{e \text{ out of } v} \ell_e =: L_v$$

The increase in flow at vertex v by L_v decreases demand by L_v .

For $v' \in V'$ with corresponding vertex $v \in V$: $d_{v'} = d_v - L_v$.

feasible circulations

Theorem (14)

Let $G = (V, E)$ be a flow network with

- for all $v \in V$: demands d_v ,
- for all $e \in E$: lower bounds ℓ_e and capacities c_e .

Let $G' = (V', E')$ be the corresponding flow network with

- for all $v' \in V'$: demands $d_{v'} = d_v - L_v$, v' corresponds to $v \in V$,
- for all $e' \in E'$: capacities $c_{e'} = c_e - \ell_e$, e' corresponds to $e \in E$.

There is a feasible circulation in G if and only if there is a feasible circulation in G' .

Proof. If and only if means $\Rightarrow$ and $\Leftarrow$.

- $\Rightarrow$ For a feasible circulation f in G , let f' in G' be $f'(e) = f(e) - \ell_e$.
- $\Leftarrow$ For a feasible circulation f' in G' , let f in G be $f(e) = f'(e) + \ell_e$.

capacity and conservation conditions for $\Rightarrow$

- $\Rightarrow$ For a feasible circulation f in G , let f' in G' be $f'(e) = f(e) - \ell_e$.

① *capacity conditions*: as f is a feasible circulation, for all $e \in E$,

$$\ell_e \leq f(e) \leq c_e \quad \Rightarrow \quad 0 \leq f'(e) \leq c_e - \ell_e, \text{ for all } e \in E'.$$

② *conservation conditions*: for all $v' \in V'$,

$$\begin{aligned} (f')^{\text{in}}(v') - (f')^{\text{out}}(v') &= \sum_{e \text{ into } v'} f'(e) - \sum_{e \text{ out of } v'} f'(e) \\ &= \sum_{e \text{ into } v'} (f(e) - \ell_e) - \sum_{e \text{ out of } v'} (f(e) - \ell_e) \\ &= L_v + \sum_{e \text{ into } v} f(e) - \sum_{e \text{ out of } v} f(e) \\ &= L_v + d_v, \text{ as } f \text{ is a feasible circulation} \\ &= L_v + d_{v'} - L_v \\ &= d_{v'}. \end{aligned}$$

capacity and conservation conditions for $\Leftarrow$

- $\Leftarrow$ For a feasible circulation f' in G' , let f in G be $f(e) = f'(e) + \ell_e$.

① *capacity conditions*: as f' is a feasible circulation, for all $e \in E'$:

$$0 \leq f'(e) \leq c_e - \ell_e \quad \Rightarrow \quad \ell_e \leq f(e) \leq c_e, \text{ for all } e \in E.$$

② *conservation conditions*: for all $v \in V$,

$$\begin{aligned} f^{\text{in}}(v) - f^{\text{out}}(v) &= \sum_{e \text{ into } v} f(e) - \sum_{e \text{ out of } v} f(e) \\ &= \sum_{e \text{ into } v} (f'(e) + \ell_e) - \sum_{e \text{ out of } v} (f'(e) + \ell_e) \\ &= L_v + \sum_{e \text{ into } v'} f'(e) - \sum_{e \text{ out of } v'} f'(e) \\ &= L_v + d_{v'}, \text{ as } f' \text{ is a feasible circulation} \\ &= L_v + d_v - L_v \\ &= d_v. \end{aligned}$$

Q.E.D.

integer valued circulations

All calculations in the proof happened with integer arithmetic.
We therefore have the following:

Corollary (15)

*Assume all demands, lower bounds and capacities are integers.
There is an integer valued feasible circulation in G if and only if
there is an integer valued feasible circulation in G' .*

This implies that the Ford-Fulkerson algorithm
will compute a feasible circulation in time $O(mD)$
for a graph with m edges and

$$D = \sum_{v \in V: d_v > 0} d_v.$$

iterative algorithms

The Ford-Fulkerson algorithm solves the max flow problem in a weighted directed graph. It is an iterative algorithm.

As the cost of the algorithm depends on the size of the numbers, its running time is pseudo polynomial.

In this lecture, we applied this algorithm

- 1 to computing the largest matching in a bipartite graph, and
- 2 to the feasibility problem in circulation with demands.

In the next lecture, we consider computing disjoint paths in graphs, the survey design problem, and airline scheduling.

Many problems can be reduced to the max flow problem.